

Research on the vortex-induced vibration and aerodynamic mitigation measures of parallel Π -shaped composite girders under aerodynamic interference effects

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ABSTRACT

This study investigated the vortex-induced vibration (VIV) characteristics and triggering mechanism of the parallel Π -shaped composite girders by wind tunnel test and Computational Fluid Dynamics (CFD) simulations. The effects and suppression mechanisms of aerodynamic mitigation measures, such as vertical stabilizer plates, horizontal plates, and fairings, on VIV were investigated. The results showed that vertical VIVs occurred in the parallel bridge at an investigated angle of attack (AOA) of 0° , -3° , and 3° . The characteristics of the VIV responses of the upstream deck are similar to those of the single bridge. However, the VIV amplitude of the downstream deck is magnified and far surpasses the allowable value under the effects of the wake vortex of the upstream deck, particularly at a positive AOA. The vertical stabilizing plate can only suppress the VIV of the upstream deck because it mitigates the vortex formation on the lower of the upstream deck. Nevertheless, the fairing has the ability to simultaneously suppress the vortices on the upper and lower of the upstream deck and minimize the influence of its wake vortices on the downstream deck. It effectively prevents the merging and distortion of vortices on the upper of the downstream deck, thereby suppressing the VIV of parallel Π -shaped composite girders. In addition, the lengthening and bilateral arrangement of the fairing can augment its ability to suppress VIV. Particularly at a -3° AOA, it successfully creates a highly stable shear layer of the upstream deck and makes the vortices on the surface of the downstream deck significantly suppressed.

1. Introduction

In recent years, owing to the robust traffic capacity, parallel bridges have been extensively constructed in large cities. These bridges include the Tacoma Narrows Bridge in the USA, the Meiko-Nishi Bridge in Japan, the Jindo Bridge in South Korea, and the Qingshuipu Bridge in China. In contrast to the single bridge, parallel bridges have the intricate aerodynamic interference effect. The aerodynamic interference effect may adversely affect the vortex-induced vibration (VIV) performance of the parallel bridge [1–3]. In fact, VIVs have been observed several times on the Jindo parallel bridge in South Korea, and a series of wind tunnel test results show that it is affected by the interference effect [4]. It is, therefore, of great importance to improve the understanding of the aerodynamic interference effect on VIV of parallel bridges and prevent its occurrence.

The investigation of the aerodynamic interference effect originated

from the study of the aerodynamic performance of tandem cylinders with circular and square cross-sections. The aerodynamic interference effect of tandem structures will involve very complex physical problems, including vortex shedding, flow separation, transition from steady to unsteady, flow-induced vibration, etc. Bentley et al. [5] studied the vortex shedding mechanism of the single and double bluff bodies, noting a significant variation between the two, which is heavily influenced by the gap's geometry of double bluff bodies. Mizushima et al. [6] studied the instability and transition of flow past tandem circular cylinders and found that there exists a specific range of gap spacing where the drag and lift coefficients exhibit a sudden change as the gap spacing is continually adjusted. Carmo et al. [7] investigated the possible flow states in the drag inversion range of the flow around tandem circular cylinders at low Reynolds numbers and carefully identified the regions of bistable flow for the first time. Rastan et al. [8] studied the effect of cylinder spacing and Reynolds number on the flow, wake transition, and

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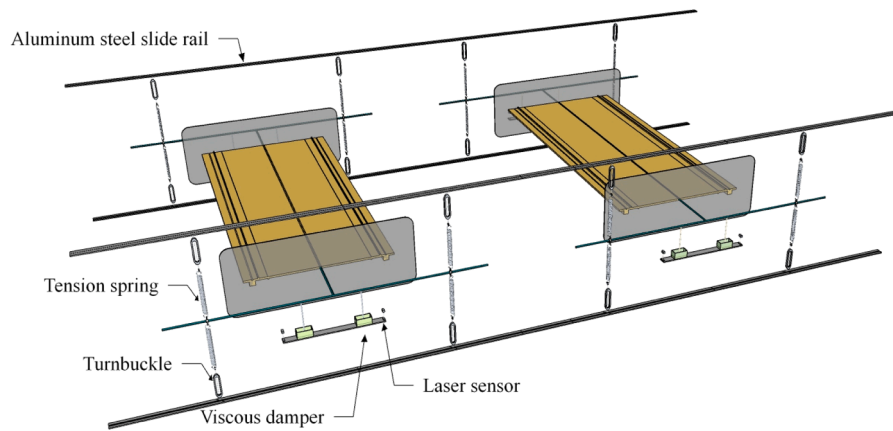


Fig. 1. Schematic diagram of the dynamic sectional testing of parallel girders.

formations for tandem circular or square cylinders and identified the borders of flows by Reynolds and spacing, including steady, unsteady, transition, slender/extended body, etc.

With the increasing number of parallel bridges designed and constructed in recent years, the research on the aerodynamic characteristics of parallel bridges has gradually increased. Wang et al. [9] researched how aerodynamic interference affects the aerostatic coefficients of parallel twin-box bridges and pointed out that the downstream deck is greatly affected due to the shielding effect. Nobuyuki [1] analyzed the VIV performance of parallel twin-box bridges by wind tunnel tests. The results indicate that the aerodynamic interference effect impacts the VIV maximum amplitude and the lock-in region of parallel bridges. Kim et al. [4] and Seo et al. [10] conducted field measurements of VIV in the Jindo parallel bridges and recreated the conditions in wind tunnel tests, revealing that the VIV was intensified by the alternating vortex created between the two parallel girders.

Spacing is the most direct factor of aerodynamic interference to parallel bridges. Some scholars studied the influence of the spacing between upstream and downstream decks on the aerodynamic interference and VIV performance of parallel bridges. Kimura et al. [11] observed that even when the spacing distance is as much as 8 times the width of the girder, the VIV performance of parallel twin-box bridges will also be greatly affected by aerodynamic interference. Park et al. [12] investigated the effect of spacing on VIV of parallel twin-box bridges by wind tunnel tests. The results showed that the spacing distance of 5–7 times the depth of the girder will cause the most significant aerodynamic interference effect on VIV response. Recently, He et al. [13] studied the aerodynamic characteristics of parallel twin-box girders through large eddy simulation and the results showed that when the spacing is 0.25–5 times the bridge width, the aerodynamic interference effect has the greatest influence on the aerodynamic coefficients. In addition to the spacing, Meng et al. [14] also found that the aerodynamic interference of parallel twin-box bridges with the same aerodynamic shape will have a considerable impact on the VIV response. Furthermore, Argentini et al. [15] observed that the upstream vibration will have a forcing effect on the downstream vibration, particularly when the frequencies of the upstream and downstream decks are in close proximity. Similarly, Park et al. [16] studied the influence of the natural frequency difference of parallel box-girders on their VIV characteristics systematically. Their research shows that the downstream VIV amplitude reaches its maximum at the same frequency.

As reviewed above, it is worth noting that the previous research on parallel bridges has mainly focused on parallel twin-box girders and analyzing how parameters like spacing affect the VIV performance of parallel bridges. Few studies have been conducted on the characteristics and mitigation of the VIV of parallel Π -shaped composite girders. Until now, Extensive studies have been carried out on the VIV mitigation

measures for a single Π -shaped girder by wind tunnel testing, numerical simulations, and field measurements [17–21]. These measures include fairing, lower central stability, an inverted L-shaped edge plate, and a lower horizontal edge plate. Although parallel bridges have greater traffic capacity, their aerodynamic configurations are more complex than single-girder ones. The existing aerodynamic measures suitable for single Π -shaped girder may be difficult to apply directly to parallel Π -shaped girders [22]. There is clearly an increasing need for more research concerning the VIV performance of parallel Π -shaped girders and related mitigation measures.

In the present study, the VIV performance and related mitigation measures of parallel Π -shaped bridges are studied via many section model wind tunnel tests. The significant VIV response of the parallel Π -shaped bridges appears at different AOA, especially for downstream decks. Different aerodynamic mitigation measures, including the fairing, vertical stabilizer, inverted L-shaped edge plate, and horizontal edge plate, were used individually and in combinations in the tests to mitigate and control the VIV responses. Furthermore, several optimization studies have been conducted on the fairing. To achieve a deeper understanding of the VIV mechanisms and the vibration suppression mechanism of aerodynamic measures, the flow structures around the original girders and the girders with aerodynamic measures were explored qualitatively by CFD technology. This study was conducted in response to the need to provide useful information and guidance for the VIV control design of parallel Π -shaped bridges.

2. Experimental and numerical setup

2.1. Experimental setup

The section model wind tunnel tests were performed in a smooth flow at the second test section of the XNJD-1 wind tunnel of Southwest Jiaotong University. The cross-sectional dimensions of the wind tunnel are 2.4 m in width, 2.0 m in height, and 16.0 m in length. The wind speed can be adjusted within a range of 0.5 to 45.0 m/s while the turbulence intensity remains below 0.5 %.

An experimental device system is developed for the purpose of conducting dynamic sectional testing on parallel girders, as illustrated in Fig. 1. The parallel girders are partitioned into upstream and downstream configurations, which are set up independently. Each model's suspension is achieved by a rigid slide rail bracket, which is securely attached to the wall of the wind tunnel. The bracket has eight linear tensile springs, enabling the models to oscillate in vertical and torsional degrees of freedom. The orchid screw is situated amidst the tensile spring and slide rail to adjust the relative height of the two girders at different AOA. The present study utilized six laser sensors positioned on both sides of the wind tunnel's inner wall. These sensors were employed



Fig. 2. Section model of parallel bridges in XNJD-1 wind tunnel.

to measure the signals of vibration displacement. To ensure a good frequency response, the system dampers were used and the tubing lengths were less than 0.2 m [23]. And the data collection procedure employed a sampling frequency of 512 Hz, accompanied by a sampling duration of 60 s

The main frames of models are predominantly constructed of wood, which is the main contribution to the structure's stiffness. Some detailed components, such as beams, are fabricated using glass fiber, enhancing structural integrity. The materials utilized for the construction of sidewalk panels, handrails, crash barriers, and maintenance vehicle tracks consist of acrylonitrile butadiene styrene. Furthermore, components of

aerodynamic configuration are also constructed from acrylonitrile butadiene styrene. To maintain a two-dimensional flow, two thin plates are affixed to both ends of the section model. Fig. 2 shows the section model mounted in the wind tunnel.

The prototype cross-section of parallel Π -shaped bridges is shown in Fig. 3; a parallel Π -shaped composite girder with a main span of 328 m was chosen as the research object. A symmetrical arrangement of twin girders on the upstream and downstream sides, with a clear gap spacing (L) of 18.16 m. And for each girder, the width (B) and height (H) are 22.75 m and 2.28 m, respectively. The selection of the geometrical scale of the section model is 1:80, which yields a blockage ratio of 0.8 %, trying to achieve the necessary blockage ratio requirements. The length of the section model is 1.41 m, so its aspect ratio meets the requirements [24].

As indicated in Table 1, the test parameters of the parallel bridge's downstream and upstream deck models remain consistent. These parameters include geometrical dimension, mass, mass moment, damping ratio, and natural frequency. The Scruton number of the sectional model is $Sc = 4\pi m \xi_v / (\rho B^2) = 9.4$, and a frequency ratio of 14.1 was selected, leading to a wind speed ratio equivalent to 5.6. Furthermore, it should be noted that the torsional damping ratio (0.32 %) is lower than that of the actual bridge. No torsional VIV occurred in the region of wind speed varied from 0 m/s to 27 m/s. Hence, this study focuses solely on the vertical VIV of parallel Π -shaped girders, independent of the torsional damping ratio.

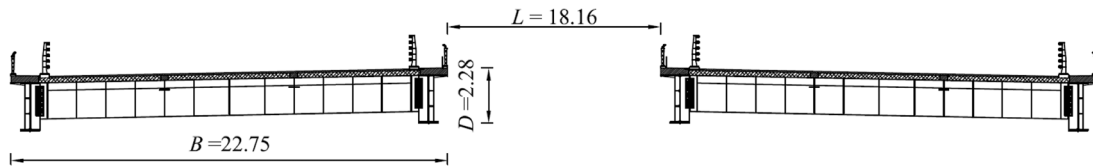


Fig. 3. Cross section of prototype parallel Π -shaped bridges (unit m).

Table 1
Parameters of wind tunnel tests.

Property	Unit	Scale ratio	Prototype	Upstream deck	Downstream deck
Length (L)	m	/	/	1.41	1.41
Height (H)	m	1/80	2.28	0.028	0.028
Width (B)	m	1/80	22.75	0.284	0.284
Mass per unit length (m)	kg/m	$1/80^2$	5.93×10^4	9.262	9.262
Mass moment of inertia per unit length (I_m)	$\text{kg}\cdot\text{m}^2/\text{m}$	$1/80^4$	3.11×10^6	0.075	0.075
Vertical frequency (f_n)	Hz	/	0.432	6.11	6.12
Torsional frequency (f_t)	Hz	/	0.892	12.34	12.36
Vertical damping ratio (ξ_h)	%	/	/	0.81	0.80
Torsional damping ratio (ξ_t)	%	/	/	0.32	0.31

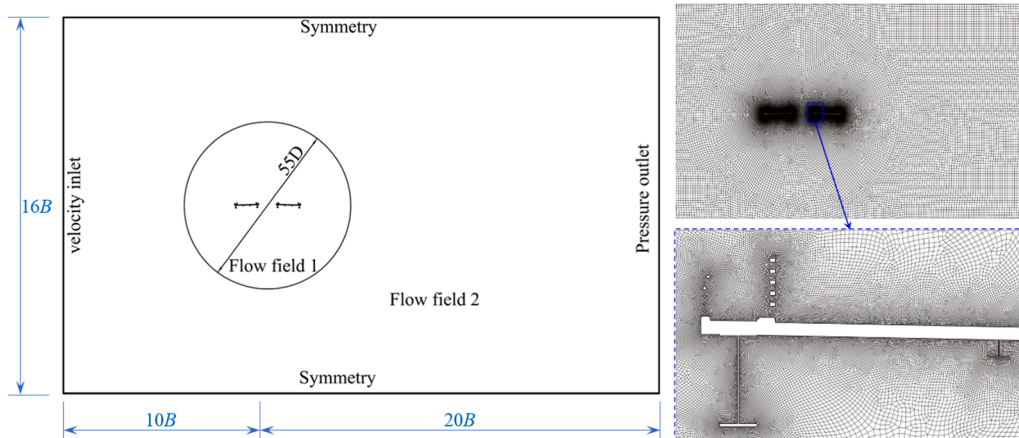


Fig. 4. Layout of computational domain and grid division diagram.

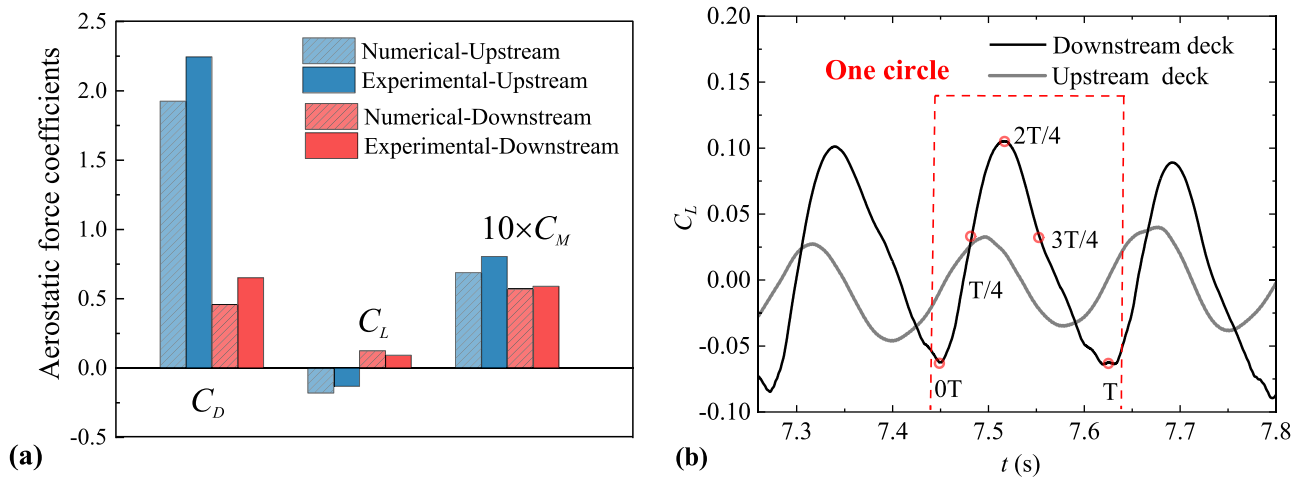


Fig. 5. Comparison of aerodynamic coefficients by numerical simulation and wind tunnel test: (a) mean value of C_D , C_L , and C_M ; (b) time-history of C_L .

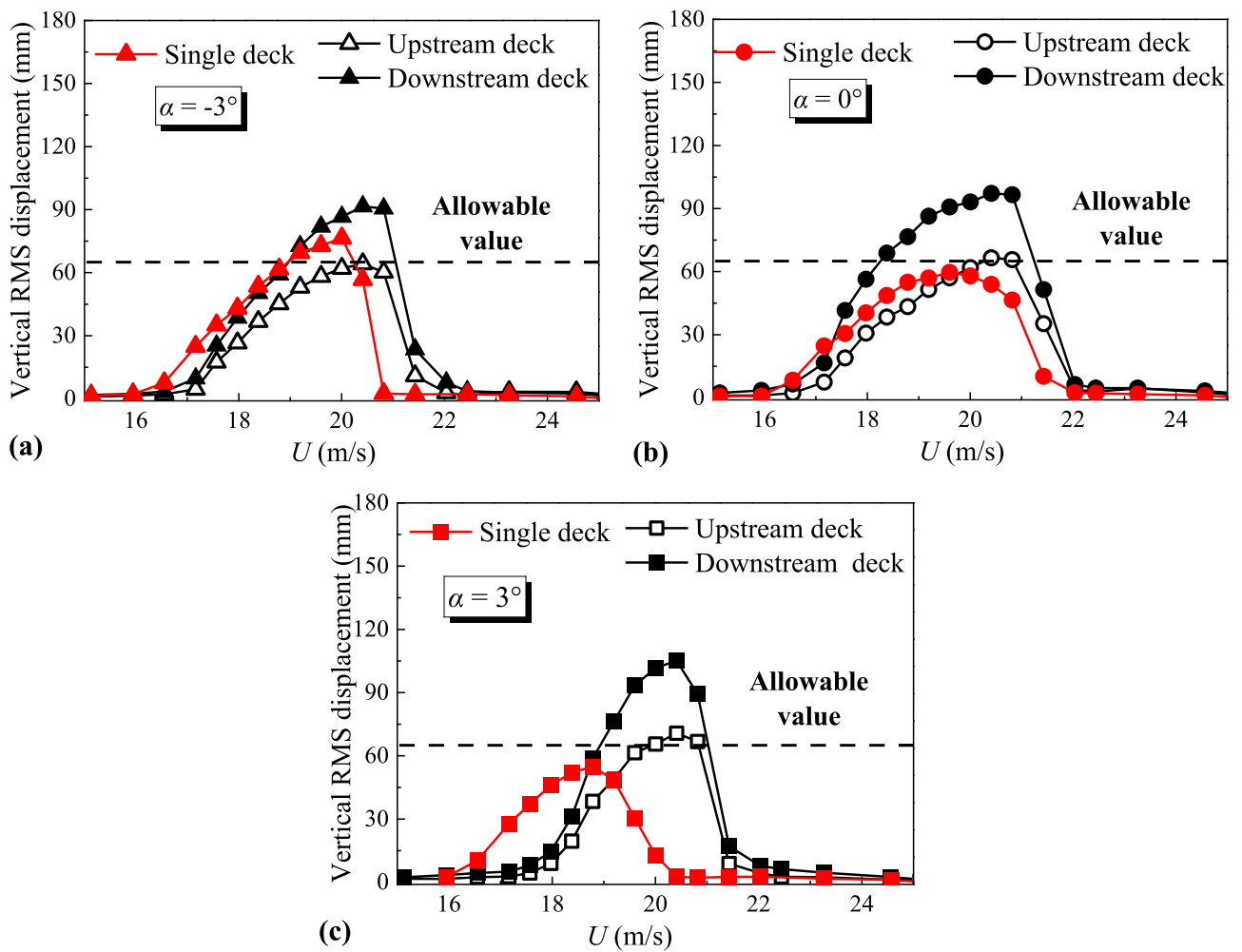


Fig. 6. Vertical VIV displacements of the original section: (a) $\alpha = -3^\circ$; (b) $\alpha = 0^\circ$; (c) $\alpha = 3^\circ$.

2.2. Numerical simulation setup

As shown in Fig. 4, In this paper, the commercial software ANSYS FLUENT is used to simulate the two-dimensional (2D) flow field around the parallel bridge section. The computational domain is set as $16B \times 30B$, where B is the width of the section. The geometric dimensions of the numerical model are consistent with those of the section model in

the physical wind tunnel tests. The multi-block meshing strategy generates high-quality and low computational complexity grids. Given the computational cost precision [19; 26], the height of the first grid layer near the cross-section is selected from the derived values to ensure that $y^+ < 1.5$ simulates the flow separation. The mesh size increase ratio from inside to outside is about 1.05. The total number of meshes is approximately 2.9×10^5 .

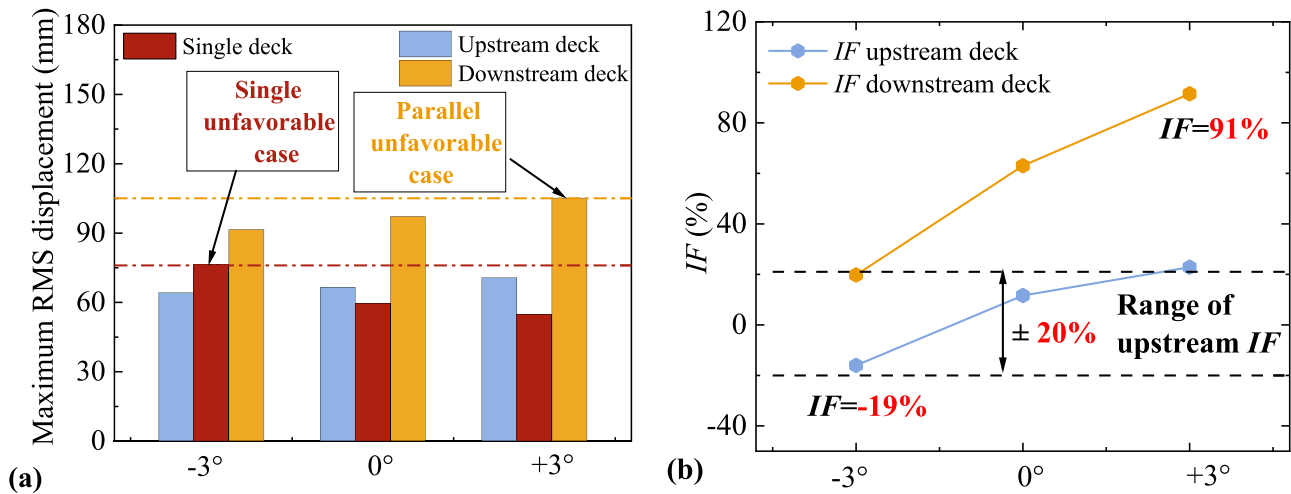


Fig. 7. Maximum VIV amplitude and IF at various AOAs: (a) maximum VIV amplitude; (b) IF .

Here, a non-steady flow field was used in the vortex shedding calculation, and the time step used in the simulations is 0.0004 s. The incoming flow speed is set to the measured flow speed in the VIV locked-in region of the parallel bridge section model, which $U = 3.2$ m/s. The uniform velocity condition without turbulence and a pressure outlet condition with zero static gauge pressure is set at the inlet and outlet boundaries, respectively. A no-slip condition is imposed for the girders, and the symmetry condition is applied on the upper and lower boundaries.

In the present numerical simulations, the URANS and SST $k-\omega$ turbulence models were used to investigate the uniform flow over the parallel Π -shaped girders. The governing equations were solved using the Fluent® package, but the options offered by Fluent® for simulation were carefully selected to achieve reliable results. For the spatial discretization, the central-differencing scheme was applied to spatial differencing of the convection term, and the second-order scheme was chosen for the pressure interpolation. The implicit second-order time-advancement scheme was chosen for temporal discretization to obtain a stable and accurate simulation. A Pressure-Implicit with Splitting Operators (PSIO) algorithm is utilized for the coupling of pressure and velocity in the numerical procedure.

To validate the present numerical method, the numerical results of the drag, lift, and pitching moment coefficients (C_D , C_L , and C_M) of the parallel bridge at AOA of 0° are compared with those from wind tunnel tests. Fig. 5(a) shows that the aerodynamic coefficients estimated by CFD match the test data well. In addition, as shown in Fig. 5(b), there is a clear periodic pattern in the numerical lift coefficients of the upstream and downstream decks, and the Strouhal number ($St = fD/U$) can be estimated to be 0.056. The aerodynamic coefficients between the numerical and experimental studies indicate that the present numerical method and grid system are applicable and can produce reasonable results.

3. VIV characteristic of original section

3.1. Wind tunnel test result

The VIV performance of the parallel bridge is initially tested at AOA of $\pm 3^\circ$, and 0° by dynamic sectional testing in uniform flow. To facilitate comparative analysis, the case of wind tunnel testing was conducted for a single bridge. Fig. 6 presents the vertical VIV amplitude of the girder. The mean flow velocity (U) and the root mean square (RMS) of the VIV displacements have been converted to the prototype values. It

should be emphasized that the vertical VIV occurred at investigated wind AOAs; no torsional VIV was observed in the test. As illustrated in Fig. 6, The parallel bridges exhibit clear vertical VIV response at investigated AOAs. It is evident that the maximum VIV amplitude of the downstream deck is considerably greater than that of the upstream deck and single girder. And Based on the Chinese Wind-resistant Design Specification for Highway Bridges [25], the allowable RMS amplitude of the vertical VIV displacements for this parallel bridge is 66 mm. It's clear that the VIV amplitude of the downstream deck greatly exceeded the allowable value. Therefore, it is necessary to optimize the VIV performance of the girder and ensure the safety of the bridge.

To further illustrate the aerodynamic interference effects and VIV characteristics of parallel bridges at different AOAs. Fig. 7(a) shows the maximum VIV amplitude for both the parallel bridge and the single bridge at various AOAs. The unfavorable case for a single bridge occurs at a negative AOA of -3° . On the contrary, interestingly, due to the aerodynamic interference effect, the unfavorable case for parallel bridges occurs at a positive AOA of 3° , where the amplitude of the downstream deck reaches the maximum value. This means that the VIV characteristics and aerodynamic interference effect of parallel bridges are sensitive to the AOA.

To quantify the aerodynamic interference effect, the interference factor is defined as $IF = M_d/M_s$ or M_u/M_s . M_d , M_u , and M_s are the VIV maximum amplitude of the downstream, upstream, and single deck. Fig. 7(b) shows the IF s of the upstream and downstream decks at different AOAs. The downstream deck experiences a notable amplification effect, particularly at an AOA of 3° ; the VIV maximum amplitude is increased by 91 % compared to a single bridge. The IF s of the upstream deck are within a range of $\pm 20\%$, which means that the aerodynamic interference effect is weak, and the VIV amplitude of the upstream deck is close to that of the single bridge. Even at -3° AOA, the IF is equal to -19% , which means that the VIV amplitude of the upstream deck is suppressed. Furthermore, compared to the single bridge, the VIV locking region of the parallel bridge is significantly delayed at -3° AOA compared to the single bridge. This evident lag may result from alteration of the attacking angle, which leads to the change of the aerodynamic structure of the girder, as well as a variation in the vortex shedding frequency.

To sum up, the aerodynamic interference effect of the parallel Π -shaped composite girders is strong, and the VIV amplitude of the downstream deck has a great amplification effect, especially at the positive AOA, which is the key case to be considered in the design of vibration control.

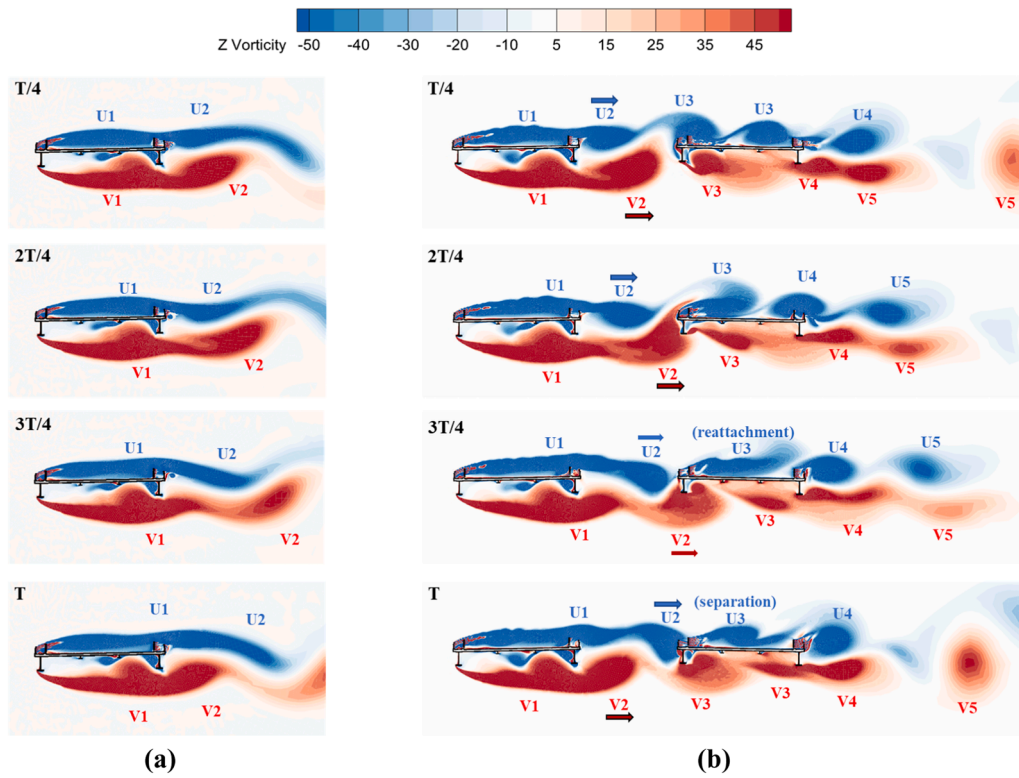


Fig. 8. Vortex evolution diagram of single bridge and parallel bridge when $\alpha = 0^\circ$: (a) single deck; (b) parallel bridge.

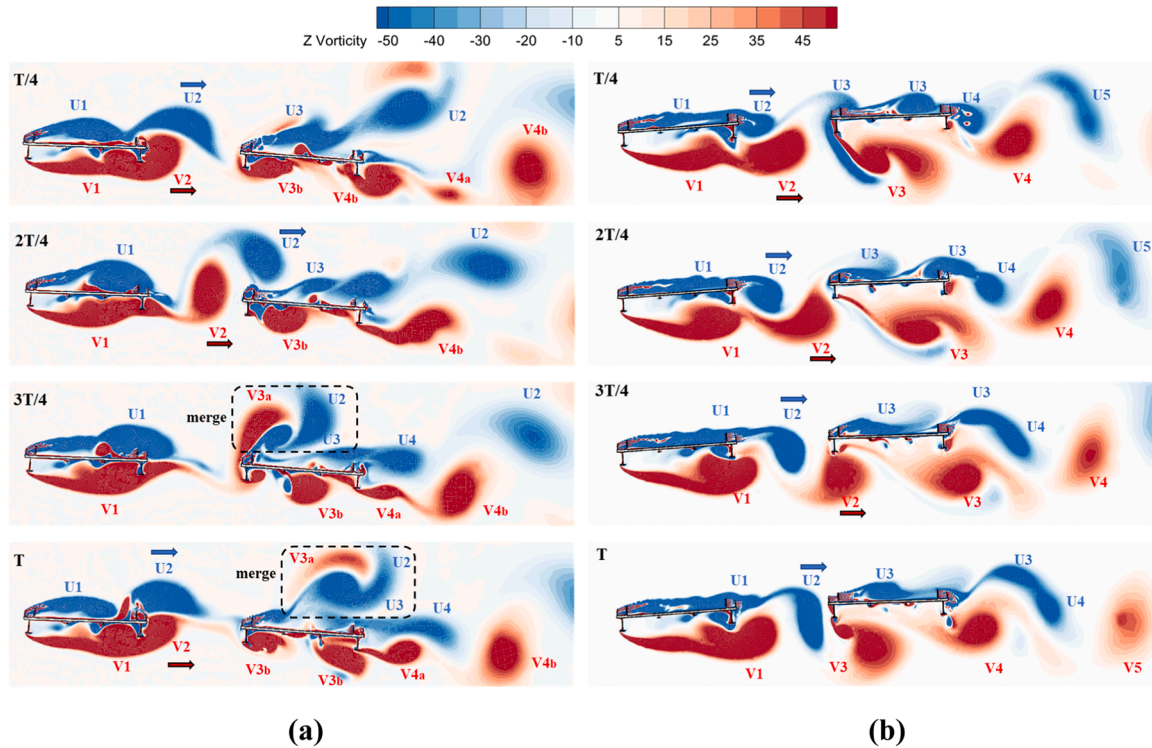


Fig. 9. Vortex evolution diagram of parallel bridges: (a) $\alpha = +3^\circ$; (b) $\alpha = -3^\circ$.

3.2. Numerical simulation result

To investigate the VIV triggering mechanism of parallel girders, the flow structures are studied by visualizing the instantaneous vortex around the single girder and parallel girders. The clockwise and

counterclockwise vorticity are depicted in blue and red, respectively.

As depicted in Fig. 8(a) and (b). The flow patterns and vortex structures surrounding the upstream deck and the single girder are highly similar to each other. It can be seen from the whole period that the incoming flow is separated at the front end of the main girder and

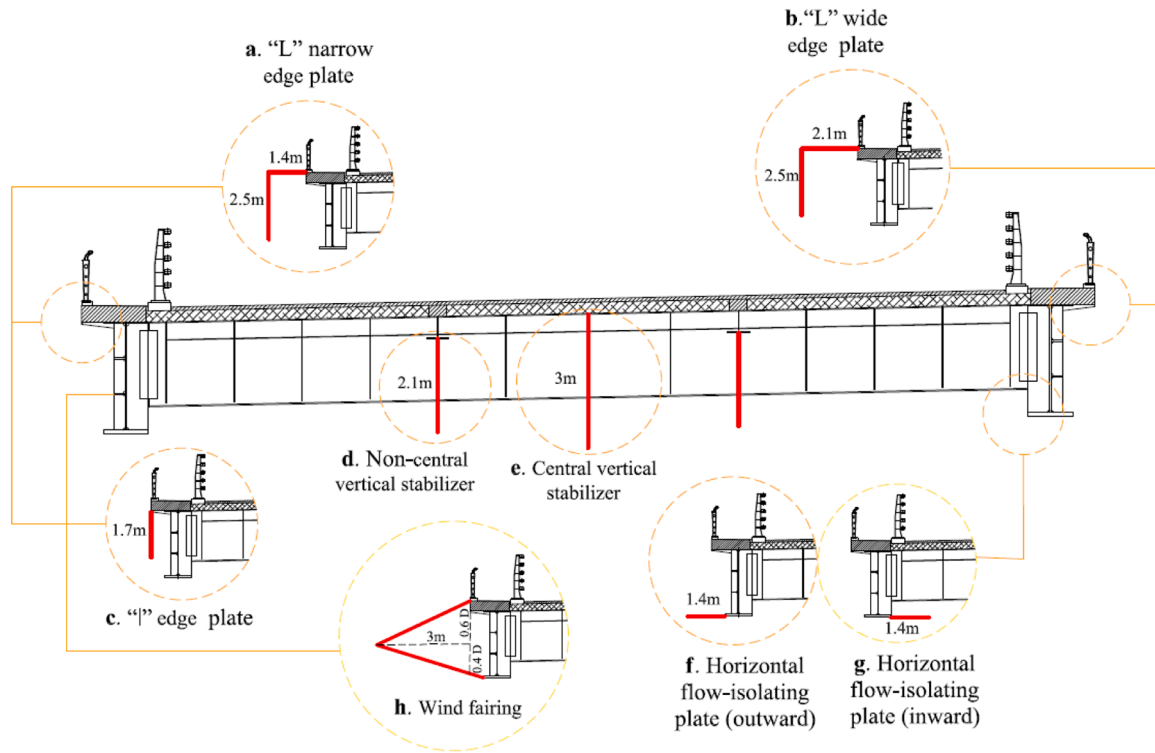


Fig. 10. Specific information of aerodynamic measures.

Table 2
Classification of 9 VIV countermeasures.

Measures	number	Details of measures
Original section	Case 0	/
Edge plates	Case 1	a. "L" narrow edge plate
	Case 2	b. "L" wide edge plate
	Case 3	c. "I" edge plate
Vertical stabilizer and horizontal plate	Case 4	d. Non-central vertical stabilizer
	Case 5	f. Horizontal flow-isolating plate (outward)
	Case 6	g. Horizontal flow-isolating plate (inward)
	Case 7	e. Central vertical stabilizer + a. "L" narrow edge plate
Combination measures and wind fairing	Case 8	e. Central vertical stabilizer + f. Horizontal plate(outward)
	Case 9	h. Wind fairing (unilateral)

develops into a large-scale vortex V1 with half bridge width at the lower part of the main girder. In contrast, a vortex U1 with a faster flow rate and smaller scale will be formed at the upper part, and then they will create an obvious shedding vortex (referred to as U2 and V2) after leaving the main girder. These processes, including the formation of the vortices U1 and V1, along with their distortion and shifting on the upper and lower surface of the girder and finally falling off at the tail to form the vortices U2 and V2, will periodically change the pressure on the surface of the girder. This leads to the vibration of a single girder or upstream girder. Similar findings have also been made in previous studies on the single Π -shaped girder [18; 21].

However, as shown in Fig. 8(b), the wake vortices U2 and V2 generated by the upstream deck drift to the downstream deck and alternately impact its leading edge, resulting in a more intricate flow field structure for the latter. There will be the formation and evolution of binary vortices above and below the downstream deck (referred to as U3 and V3 vortices), especially the vortex U3, which will undergo obvious periodic reattachment ($t = 3/4 T$) and separation ($t = 4/4 T$) as it passes through the upper surface. Finally, they form stable 'Karman

vortex streets' (called vortices U4, V4, U5, V5) at the tail. Consequently, there are substantial periodic variations in pressure on the surfaces of the downstream deck, which finally causes a large increase in the VIV amplitude on the downstream deck. Qin et al. [26] investigation of the flow-induced vibrations of two tandem circular cylinders and found a similar trigger mechanism, which reattachment, detachment, and switching of the wake vortices of gap shear layers result in largely positive work on the downstream cylinder.

The VIV characteristics of parallel bridges are sensitive to the AOA. Therefore, it is necessary to analyze the VIV triggering mechanism of the main girder at different AOAs to provide a reference for the design of suppression measures.

As shown in Fig. 9(a), when $\alpha = 3^\circ$, the upper and lower regions of the upstream deck provide sufficient space for the formation and drift of the vortex U1 and the vortex V1, respectively. As a result, the wake vortices V2 and U2 develop with great intensity. The fully developed wake vortices periodically collide with the edge of the downstream deck, affecting the formation of the shear layer surrounding it, resulting in a more intricate vortex structure on the surface of the downstream deck. Specifically, as shown in Fig. 9(a) when $t = 3 T / 4$, the merging of vortices (U2, U3, V3a) occurs in the upper region of the downstream deck, where U2 is the wake vortex that drifts to the upper part of the downstream deck, V3a is a vortex formed when the wake vortex V2 impacts the leading edge of the downstream deck and separates towards the upper part, and U3 is the vortex of the upper part itself. This results in a substantial variation in lift, and a more severe VIV response at AOA of 3° , compared with the case at 0° AOA.

As shown in Fig. 9(b), when $\alpha = -3^\circ$, unlike the case at $\alpha = 3^\circ$, the shear layer on the upper surface of the upstream deck is stable and does not show any clear separation vortices, which means that the vortex U1 has vanished. The shear layer instability exclusively occurs in the lower part, leading to the formation of the vortex V1. Of course, the pressure fluctuation on the upstream deck is reduced at $\alpha = -3^\circ$, resulting in the decrease of VIV amplitude. As a result of the change in the AOA and the

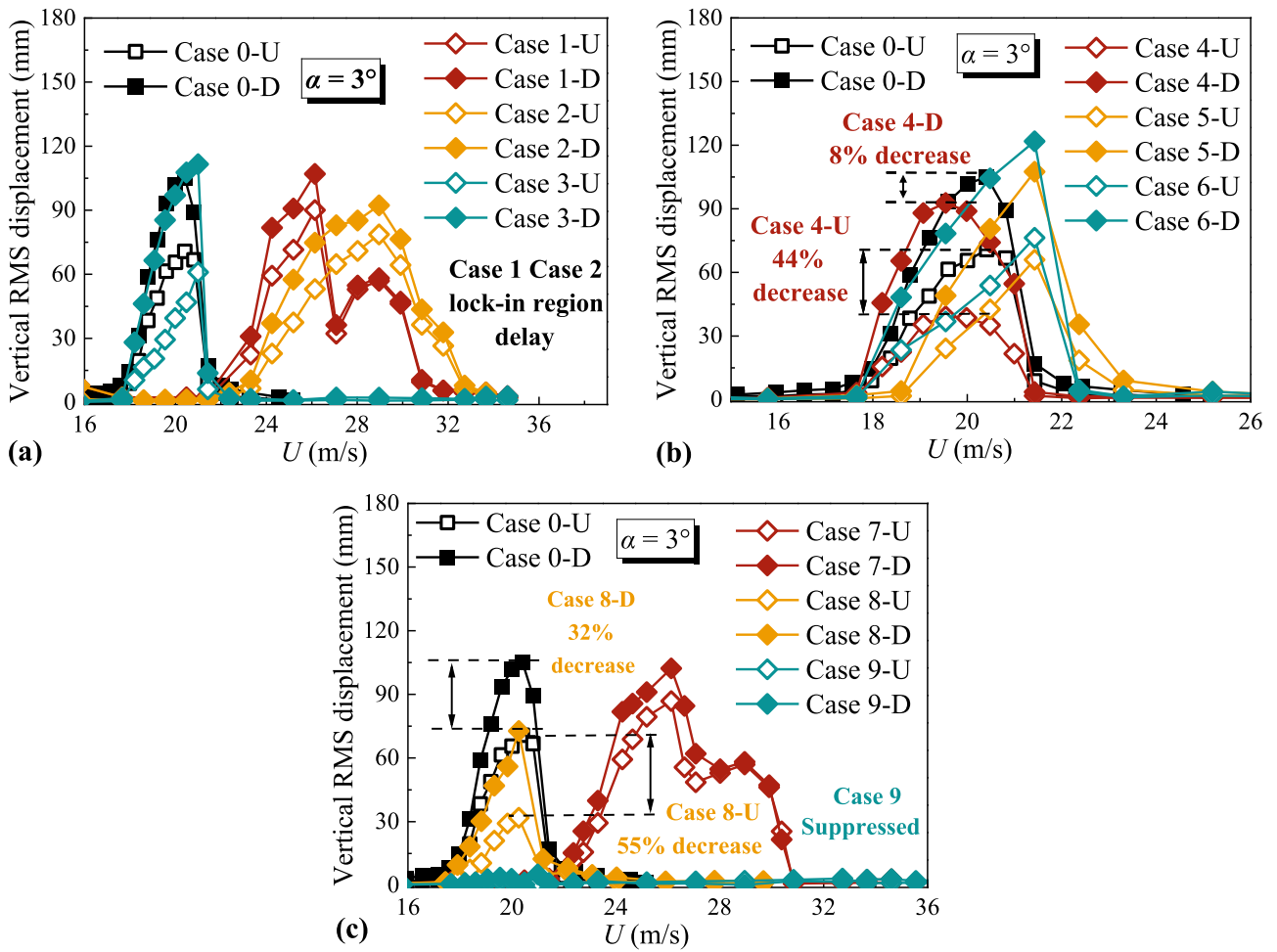
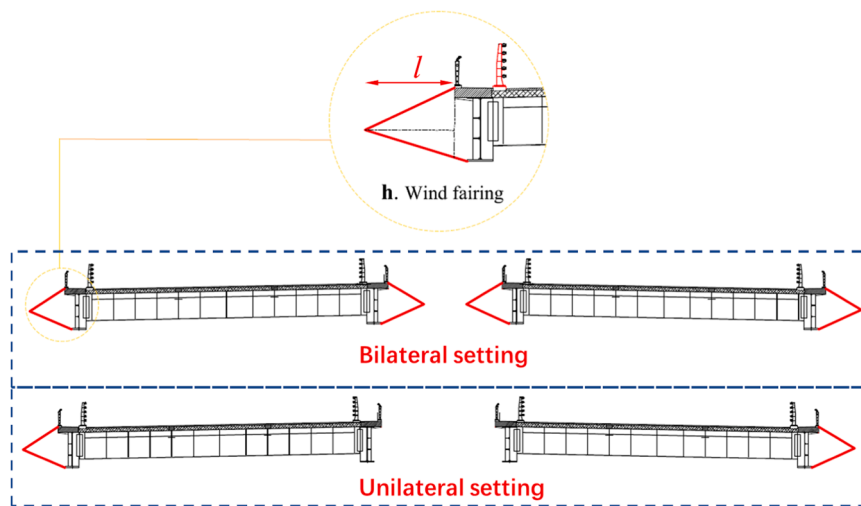
Fig. 11. Test results of VIV countermeasures at $\alpha = 3^\circ$.

Fig. 12. Optimization scheme of fairings.

flow pattern in the gap, the vortex U3 in the upper part of the downstream deck is weakened, while the vortex V3 in the lower part is amplified but further far away from the surface of the downstream deck. In fact, this will somewhat decrease the total fluctuation of pressure on the downstream deck. Compared to the case when $\alpha = 3^\circ$ or 0° , this is responsible for the minor decrease in the VIV amplitude when the $\alpha = -3^\circ$.

4. VIV mitigation countermeasure research

4.1. Preliminary countermeasure research

The parallel bridge and the twin box girders share a highly similar structure. The VIV of the twin box girders can be controlled by installing the grid plate at the gap[27]. Regrettably, for the parallel bridges, the

Table 3
Optimization schemes and VIV suppression results of fairings.

Number	l (m)	Setting mode	AOA (VIV completely suppressed)
Case 9	3	Unilateral	$3^\circ, 0^\circ$
Case 10	3	Bilateral	$3^\circ, 0^\circ$
Case 11	3.5	Bilateral	$3^\circ, 0^\circ, -3^\circ$

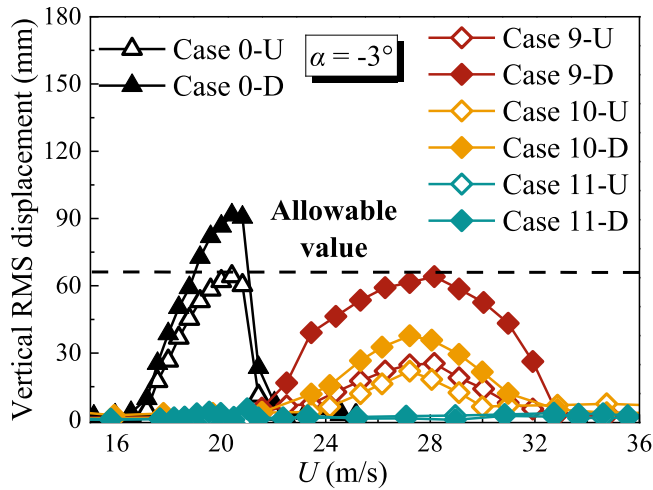


Fig. 13. Test results of VIV countermeasures of fairings at $\alpha = -3^\circ$.

upstream and downstream decks are independent of each other, and implementing such designs is not possible. Based on the aforementioned analysis of the VIV mechanism, it can be known that the wake vortex of the upstream deck is not only the direct cause of the VIV of the upstream deck but also the indirect cause of the VIV of the downstream deck. Similarly, Argentini et al. [15] also point out that the vortex vibration of the upstream deck has a forcing effect on the downstream deck. Therefore, the vibration control design of parallel bridges can be attempted by controlling the VIV of the upstream deck and reducing its wake vortex.

Hence, for the vibration control of parallel Π -shaped girders, it is still imperative to consult the prior research findings on some aerodynamic techniques for controlling VIV of a single Π -shaped girder, including such as the fairing, lower central stability, inverted L-shaped edge plate, and lower horizontal edge plate and so on [17–20]. In this study, A series of aerodynamic measures are proposed to try to mitigate the VIVs of the

parallel Π -shaped girders. Fig. 10 and Table 2 display specifics of the mitigating measures, where the original section is defined as Case 0, and the remaining measures are categorized into three groups:

- (1) Edge plates arranged at the upper end of the windward side: Case 1–3;
- (2) Horizontal plates and vertical stabilizer at the lower part of the girder: Case 4–6;
- (3) Combination measures and wind fairing: Case 7–9.

The most unfavorable wind attack angle of 3° was selected to conduct the VIV optimization tests, and the measured RMS of the VIV displacements are displayed in Fig. 11. To facilitate comparative analysis, the suffixes '-U' and '-D' are defined to represent upstream and downstream deck respectively. For instance, 'Case 1-U' represents the upstream deck of Case 1.

The main purpose of Cases 1–3 is to investigate the VIV mitigation of edge plates. As shown in Fig. 11 (a), the narrow and wide "L" edge plates (Case 1 and Case 2) both significantly increase the VIV onset wind speed but have no inhibitory effect on the VIV amplitude and even increase the maximum VIV amplitude of upstream deck as shown in Case 1-U and Case 2-U. In Case 3, the "I" edge plate also cannot suppress the VIV; the VIV amplitude approximates that of the original section.

The VIV mitigation effects of the horizontal plate and vertical stabilizer at the lower part of the girder were studied, and testing results are shown in Fig. 11(b) for Cases 4–6. Setting a non-central vertical stabilizing plate (Case 4) has some inhibitory effects, and the maximum VIV amplitudes of the upstream decks are reduced by 44 %. However, the maximum VIV amplitude of the downstream deck in Case 4 is only reduced by 8 % and is still 92 mm, which exceeds the allowable amplitude of 66 mm. In addition, the VIV mitigation effect of the outward and inward horizontal plates (Case 5 and Case 6) is worse, and their maximum VIV amplitudes are even enlarged compared with the original section.

The test results for VIV mitigation effects of the combination measures are shown in Fig. 11(c). The VIV mitigation effect of Case 7 with the central stabilizer and "L" edge plates is essentially identical to the "L" edge plates (Case 1 and Case 2), which has no inhibitory effect on the VIV amplitude but only delays the VIV lock-in region. When the combination measures Case 8 with a central stabilizer and the outward horizontal plate was used, the VIV of the upstream deck is considerably suppressed, with a reduction in maximum VIV amplitude of 55 %, while the maximum VIV amplitude of the downstream deck is reduced by 32 %, which is still higher than the allowable value. This suggests that

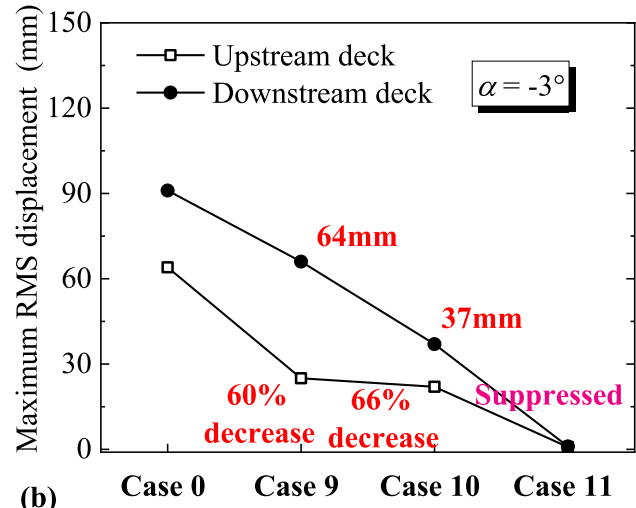
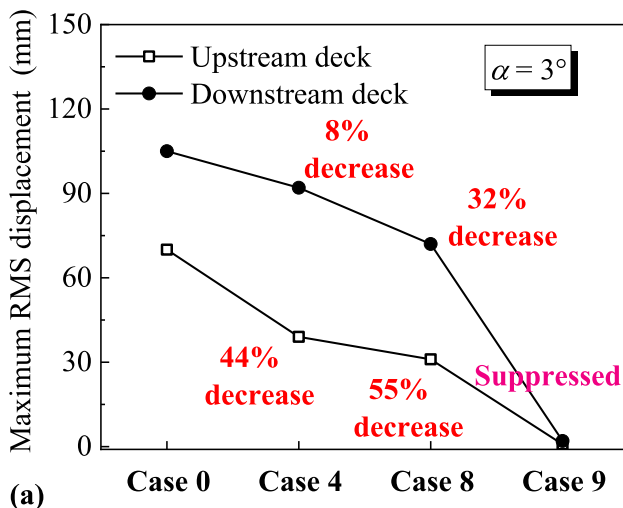


Fig. 14. Maximum amplitude of VIV countermeasures: (a) $\alpha = 3^\circ$; (b) $\alpha = -3^\circ$.

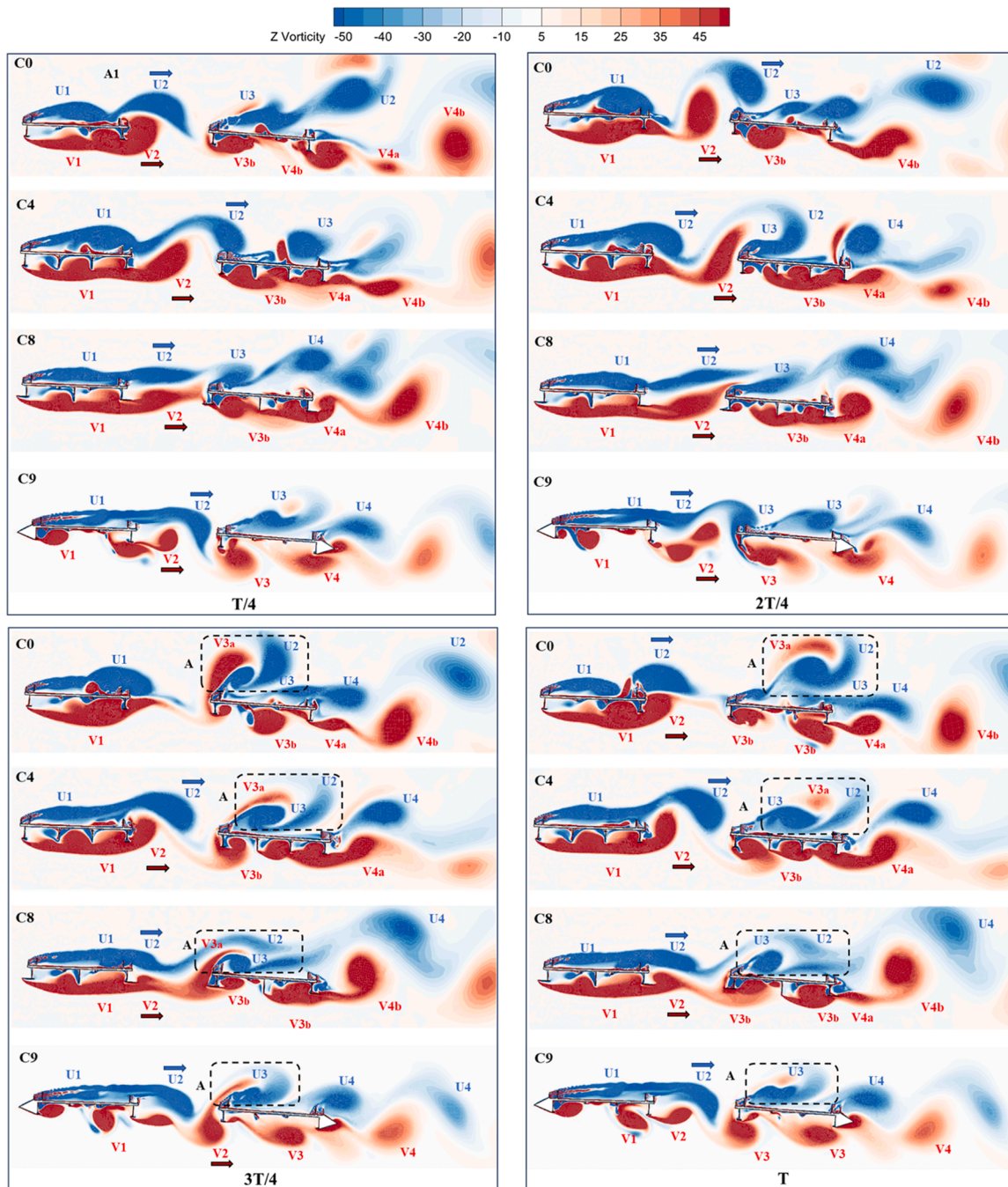


Fig. 15. Vortex evolution diagram of measures sections at 3° AOA.

the overall VIV suppression effect of the combination measure of Case 8 is not satisfactory.

More importantly, in Case 9, as shown in Fig. 11(c), it's obvious to see that setting the unilateral fairing at the windward side of the section has the best result, entirely suppressing the VIV of upstream and downstream decks at AOA of 3°. It needs to be mentioned that the measures of Case 1–8 are all arranged on both sides, that is, in the bilateral setting; only Case 9 is the unilateral setting for fairing. In conclusion, when $\alpha = 3^\circ$, only the unilateral fairing (Case 9) can effectively suppress its VIV. Therefore, the next step will be to conduct more in-depth optimization research on fairing at all AOAs.

4.2. Countermeasure optimization

To further investigate the optimization research for the fairing, the new forms of fairings (Case 10 and Case 11) are added by changing the arrangement and the horizontal length l of fairings. The schematic diagram of bilateral or unilateral setting and the size setting of fairings are shown in Fig. 12 and Table 3.

The VIV mitigation effect of each fairing at all AOAs will be tested by wind tunnel test. As shown in the rightmost column of Table 3, all the fairings in Case 9–11 can completely mitigate the VIV of the parallel Π -shaped composite girder bridge at 0° or 3° AOA. But, at an AOA of -3° , only the fairing with lengthening and bilateral setting (Case 11) can fully suppress the VIV. This means that there is still a certain VIV response in Case 9 and Case 10 at AOA of -3° .

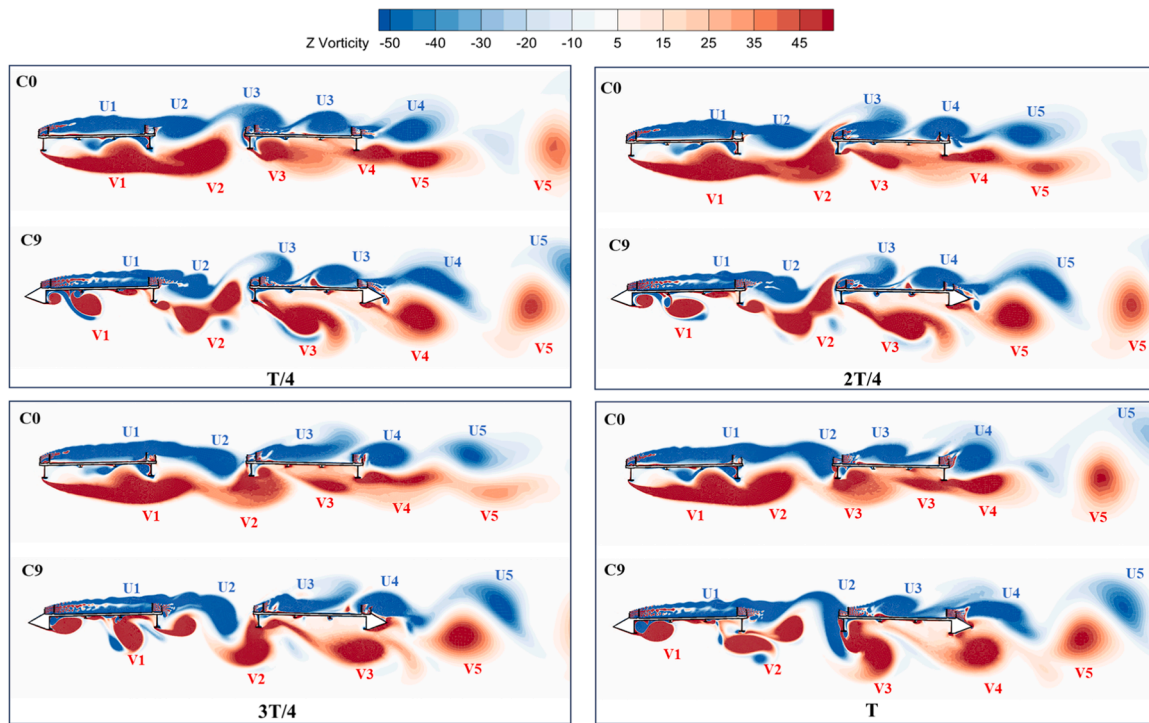


Fig. 16. Vortex evolution diagram of measures sections at 0° AOA.

Fig. 13 presents the results of the VIV response RMS at AOA of -3° . Firstly, there are VIV response in Case 9 and Case 10, and compared the original section, the VIV lock-in region are significantly delayed. In addition, different from the fairing with lengthening and bilateral setting (Case 11), which can completely suppress VIV, the fairing with bilateral setting (Case 10) reduces the maximum VIV amplitude by 60 %, while, the suppression effect of the fairing with unilateral setting (Case 9) is relatively weak, the maximum VIV amplitude is reduced by 30 %.

It is worth noting that although there is a certain VIV response at -3° AOA in Case 9 and Case 10, compared the original section, its severe VIV basically occurs after 25 m/s, and the maximum VIV amplitude is also reduced and within the allowable value. Hence, it can be said that all fairings have a good suppression effect for the VIV of the parallel II-shaped composite girder. Particularly, for the fairing with a unilateral setting and without lengthening (Case 9), it has more advantages in economy and aesthetics in real construction.

4.3. VIV mitigation mechanism

To further reveal the VIV mitigation mechanism, this paper analyzes the flow patterns of the main girder of the parallel bridge in Case 4, Case 8, and Case 9 when $\alpha = 3^\circ$, Case 9 when $\alpha = 0^\circ$, Case 9, Case 10 and Case 11 when $\alpha = -3^\circ$. In addition, Fig. 14 presents the maximum VIV amplitude of upstream and downstream decks for each Case when $\alpha = -3^\circ$ and 3° . This allows for a clear comparison of the difference in the vibration control effect of each measure, providing a reference for analyzing the relevant measures' VIV suppression mechanisms.

The vortex evolution diagrams of main girder in Case 4, Case 8, Case 9 and original section Case 0 at AOA of 3° are shown in Fig. 15. For the upstream deck, the non-central vertical stabilizer and central vertical stabilizer setting at the lower part of the girder (Case 4 and Case 8) can disperse the large-scale vortex V1 on lower part of girder, resulting in a partial reduction of VIV response for upstream deck. However, the vertical stabilizer has little effect on the shear layer on the upper of the upstream deck, so like the original section, there is a developed vortex U1 on the upper of a girder, and together with obvious wake vortices U2

and V2 at the tail. It is worth mentioning that compared to Case 4, the action of the outward horizontal plate of Case 8 results in a minor decrease in the strength of U1, U2, and V2, which enhances the VIV suppression effect for the upstream deck, reducing the maximum VIV amplitude by 55 %, while Case 4 is 44 %.

However, unilateral fairing (Case 9) can not only greatly reduce the strength and size of the lower vortex V1 but also basically eliminate the upper vortex U1. The wake vortex U2 and V2 are also significantly reduced, thereby completely suppressing the VIV of the upstream deck.

For the downstream deck, when $t = T/4$ to $t = 2T/4$, the intensity of the wake vortex U2 and V2 at the gap is weakened from measure Case 4 to Case 8 and then to Case 9, resulting in different effects on the downstream deck. From $t = 3T/4$ to $t = T$, these different effects are mostly observed in that the strength of the vortex on the upper of the downstream deck (region A) is progressively diminished. Specifically, similar original section Case 0, there are obvious vortexes (U2, U3, V3a) merging in region A in Case 4, so the non-central vertical stabilizer has less suppression effect on the VIV of the downstream deck, that is, as shown in Fig. 14(a) only 8 % of the maximum VIV amplitude is reduced. Implementing Case 8 with central vertical stabilizer and outward horizontal plate partially diminishes the intricate vortexes in region A, but with little impact, resulting in a mere 32 % reduction in the maximum VIV amplitude.

However, the unilateral fairing (Case 9) ensures that there is no merging of vortexes in region A, and the vortexes U2 and V3a essentially vanish. There is only a low-intensity vortex U3 that dissipates rapidly. In addition, the unilateral fairing effectively reduces the size of the vortex V3b on the lower of the downstream deck, moving it further away from the main girder. Therefore, the unilateral fairing can well suppress the VIV of the downstream deck. Dong et al. [28] studied the flow around two tandem rectangular cylinders with various single-side fairings and found a similar mitigation mechanism.

In summary, the vertical stabilizer can weaken the vortex on the lower part of the upstream deck, which can partially suppress the VIV of the upstream deck. But it has little effect on the vortex on the upper part of the upstream deck and the wake vortex, which makes the downstream deck still have significant VIV under the influence of the wake vortex.

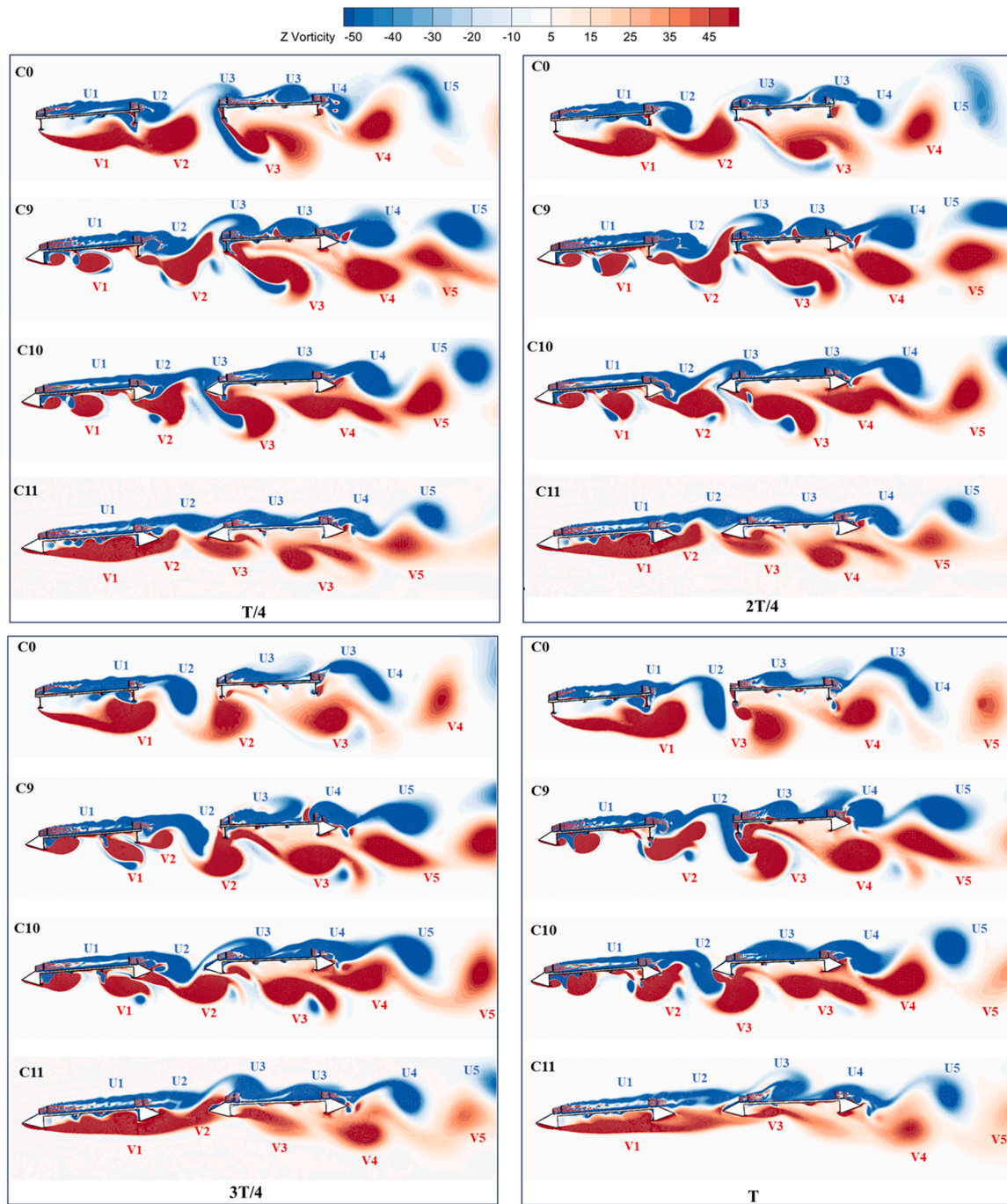


Fig. 17. Vortex evolution diagram of measures sections at -3° AOA.

But the unilateral fairing can well weaken the vortex at the upper and lower parts of the upstream deck as well as the wake vortex at the tail, resulting in reduced vorticity around the downstream deck, especially the vortex at the upper part of the main girder, thus suppressing the VIV of the parallel Π -shaped composite girders bridge at 3° AOA.

At AOA of 0° , like the case of 3° AOA, as shown in Fig. 16, throughout the entire period, compared to the original section Case 0, the unilateral fairing (Case 9) makes the large-scale vortex V1 on the lower of the upstream deck transform into a smaller-scale vortex. The shear layer on the upper region remains stable, meaning that the vortex U1 disappears and the strength of the wake vortices U2 and V2 is significantly diminished. For the downstream deck, the vortex U3 on the upper of the girder becomes weaker, the vortex V3 on the lower is far away from the surface of the main girder, and there is an absence of a stable 'Carmen

vortex street' at the tail. Consequently, the unilateral fairing reduces the pressure fluctuations on the surface of the upstream and downstream decks. This reduction effectively suppresses the vortex vibration at 0° AOA.

The vortex evolution diagrams of Case 9, Case 10, Case 11, and Case 0 at AOA of -3° are shown in Fig. 17. For the upstream deck, compared with the original section Case 0, the fairing with unilateral and bilateral setting (Case 9 and Case 10) both effectively decrease the size of vortex V1 on lower of the main girder, which make the maximum VIV amplitude of upstream deck is decreased by 60 % and 66 %, respectively, according to Fig. 14(b). Unfortunately, the unilateral fairing (Case 9) amplifies the vortex U3 on the upper of the downstream deck, which causes the downstream deck to still have a significant VIV response. As compared to the unilateral fairing, the bilateral fairing (Case 10) adds a

fairing at the windward front of the downstream deck, which results in a more stable shear layer on the upper of the girder and a notable decrease in the strength of the vortex U3. Hence, there is a decrease in the maximum VIV amplitude for the downstream deck to 37 mm in Case 10, while in Case 9, it is 64 mm.

However, the lengthened and bilateral fairing (Case 11) makes the separation of the incoming flow occur at an earlier stage. This results in improved stability of the shear layer in the upper and lower parts of the upstream deck; that is, the vortex U1 and V1 are mostly eliminated. Additionally, not only does the vortex U3 on the upper of the downstream deck become smaller and smaller, but the vortex V3 on the lower is obviously suppressed. Hence, as illustrated in Fig. 14(b), the lengthened and bilateral fairing effectively mitigates the VIV of the parallel Π -shaped composite girders bridge at an AOA of -3° .

5. Conclusions

In the current study, the VIV performances and mitigation mechanisms of parallel Π -shaped composite girders under different AOAs were studied through a series of section model wind tunnel tests and CFD simulations. Based on the results, the VIV characteristics and mitigation mechanisms of the parallel Π -shaped girders can be concluded as follows:

- There is a significant vertical VIV response of parallel Π -shaped composite girders. The upstream deck is less affected by the aerodynamic interference effect, and the VIV characteristics are close to that of the single bridge. However, under the influence of the wake vortex of the upstream deck, the unstable double vortices form on the downstream deck, especially when there is a complex vortex merging on the upper of the downstream deck at the AOA of 3° . These phenomena cause significant periodic fluctuations in the surface pressure of the downstream deck so that the vertical VIV amplitude of the downstream deck is significantly higher than that of a single bridge and far more than the allowable value.
- At an AOA of 3° , various edge plates and horizontal plates can't suppress the VIV of parallel Π -shaped composite girders. The vertical stabilizer can only weaken the vortex on the lower of the upstream deck so that the VIV performance of the upstream deck is improved, but the VIV of the downstream deck is still severe. However, unilateral fairing can suppress the vortices on the upper and lower of the upstream deck and minimize the influence of its wake vortices on the downstream deck, especially alleviating the complex vortex on the upper of the downstream deck so that the unilateral fairing completely suppresses the VIV of parallel Π -shaped composite girders. In addition, at AOA of 0° the unilateral fairing can also completely suppress the VIV and has a similar mitigation mechanism.
- At AOA of -3° , although unilateral fairing makes the VIV amplitude of parallel Π -shaped composite girders suppressed to an allowable value and the lock-in region is delayed, on the other hand, it amplifies the vortices on the upper of the downstream deck, which causes the downstream deck still have a VIV response. The lengthened and bilateral setting for fairing can enhance the ability of the fairing to suppress VIV. It effectively ensures the stability of the shear layer of the upstream deck and limits the development of the wake, so that also significantly suppresses the vortices on the downstream deck, which completely controls VIV at AOA of -3° .

CRedit authorship contribution statement

Chuan Qin: Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Ming Li:** Writing – review & editing, Supervision, Data curation. **Mingshui Li:** Writing – review & editing, Validation, Project administration.

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References

- [1] Akihiro H, Naruhito S, Masaru M, Yoichi F, Kazuo S, Nobuyuki S. Aerodynamic stability of Kansai International Airport access bridge. *J Wind Eng Ind Aerodyn* 1993;49:533–42.
- [2] Xu S, Fang G, Øiseth O, Zhao L, Ge Y. Experimental study on distributed aerodynamic forces of parallel box girders with various slot width ratios and aerodynamic countermeasures during vortex-induced vibration. *J Wind Eng Ind Aerodyn* 2023;240:105493.
- [3] Qin C, Zhou Q, Wu W, Li M. Vortex-induced vibration characteristics of parallel Π -shaped composite girders. *J Wind Eng Ind Aerodyn* 2024;252:105839.
- [4] Kim S-J, Kim H-K, Calmer R, Park J, Kim GS, Lee DK. Operational field monitoring of interactive vortex-induced vibrations between two parallel cable-stayed bridges. *J Wind Eng Ind Aerodyn* 2013;123:143–54.
- [5] Bentley JP, Mudd JW. Vortex shedding mechanisms in single and dual bluff bodies. *Flow Meas Instrum* 2003;14:23–31.
- [6] Mizushima J, Suehiro N. Instability and transition of flow past two tandem circular cylinders. *Phys Fluids* 2005;17.
- [7] Carmo BS, Meneghini JR, Sherwin SJ. Possible states in the flow around two circular cylinders in tandem with separations in the vicinity of the drag inversion spacing. *Phys Fluids* 2010;22.
- [8] Rastan MR, Alam MM. Transition of wake flows past two circular or square cylinders in tandem. *Phys Fluids* 2021;33.
- [9] Wang J, Liu JX. Test research on aerodynamic interference effect on aerostatic coefficients of main beam in parallel bridge. *Appl Mech Mater* 2012;178–181: 2131–4.
- [10] Seo J-W, Kim H-K, Park J, Kim K-T, Kim G-N. Interference effect on vortex-induced vibration in a parallel twin cable-stayed bridge. *J Wind Eng Ind Aerodyn* 2013;116: 7–20.
- [11] Kimura K, Shima K, Sano K, Kubo Y, Kato K, Ukon H. Effects of separation distance on wind-induced response of parallel box girders. *J Wind Eng Ind Aerodyn* 2008; 96:954–62.
- [12] Park J, Kim S, Kim H-K. Effect of gap distance on vortex-induced vibration in two parallel cable-stayed bridges. *J Wind Eng Ind Aerodyn* 2017;162:35–44.
- [13] He X, Kang X, Yan L, Flay RGJ, Ren P, Wu T. Numerical investigation of flow structures and aerodynamic interference around stationary parallel box girders. *J Wind Eng Ind Aerodyn* 2021;215.
- [14] Meng X, Zhu L, Guo Z. Aerodynamic interference effects and mitigation measures on vortex-induced vibrations of two adjacent cable-stayed bridges. *Front Archit Civ Eng China* 2011;5:510–7.
- [15] Argentin T, Rocchi D, Zasso A. Aerodynamic interference and vortex-induced vibrations on parallel bridges: the Ewijk bridge during different stages of refurbishment. *J Wind Eng Ind Aerodyn* 2015;147:276–82.
- [16] Park J, Kim H-K. Effect of the relative differences in the natural frequencies of parallel cable-stayed bridges during interactive vortex-induced vibration. *J Wind Eng Ind Aerodyn* 2017;171:330–41.
- [17] Zhang T, Sun Y, Li M, Yang X. Experimental and numerical studies on the vortex-induced vibration of two-box edge girder for cable-stayed bridges. *J Wind Eng Ind Aerodyn* 2020;206.
- [18] Bai H, Li R, Xu G, Kareem A. Aerodynamic performance of Π -shaped composite deck cable-stayed bridges including VIV mitigation measures. *J Wind Eng Ind Aerodyn* 2021;208.
- [19] Yoshinobu K, Kenji S, Eiki Y, Kusuo K, Yuzo O, Takashi K. Improvement of aeroelastic instability of shallow π section. *J Wind Eng Ind Aerodyn* 2001;89:14.
- [20] Li J, Wu P, Hao J, Pan H. Experimental and numerical studies on the two “lock-in” regions characteristic of vertical vortex-induced vibration of Π -shaped composite bridge deck. *J Wind Eng Ind Aerodyn* 2022;228.
- [21] Li M, Lin Z, Dai R, Wang Y, Li M. Effects of wind barriers on the aerodynamic forces and driving safety of vehicles on a wide bridge girder at various angles of attack. *Phys Fluids* 2024;36.
- [22] Stoyanoff S, Dallaire PO, Taylor ZJ, Larose GL. Aerodynamic problems of parallel deck bridges. *Bridge Struct* 2019;15:139–47.
- [23] Sun Y, Li M, Li M, Liao H. Spanwise correlation of vortex-induced forces on typical bluff bodies. *J Wind Eng Ind Aerodyn* 2019;189:186–97.
- [24] Li M, Li M, Zhong Y, Luo N. Buffeting response evaluation of long-span bridges with emphasis on the three-dimensional effects of gusty winds. *J Sound Vib* 2019; 439:156–72.
- [25] Design, C.C.H.P.A. Wind-resistant design specification for highway bridges; 2018, p. 114P, A114.
- [26] Qin B, Alam MM, Zhou Y. Free vibrations of two tandem elastically mounted cylinders in crossflow. *J Fluid Mech* 2018;861:349–81.
- [27] Ma CM, Wang JX, Li QS, Qin H, Liao HL. Vortex-induced vibration performance and suppression mechanism for a long suspension bridge with wide twin-box girder. *J Struct Eng* 2018;144.
- [28] Dong H, Shi D, Du X. Flow around single and two tandem rectangular cylinders with various single-side fairings. *Phys Fluids* 2024;36.